# A v23a N$^3$LL Fixed-Target Drell–Yan TMD Extraction: Perturbative Backend, Data Treatment, Replica Uncertainties, and $b_T/k_T$ Representations

Internal note for the `bT-TMD` analysis

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**Abstract**

This note documents the present fixed-target Drell–Yan (DY) transverse-momentum-dependent parton distribution function (TMDPDF) extraction. The result uses E288, E605, and E772 data, the v22 full perturbative backend with a N$^3$LL-accurate resummed $W$ term (implemented in the code as `resum-order n3llp`), a corrected low-$Q$ fixed-target data table, explicit normalization priors, a controlled point-to-point uncertainty sensitivity for E772 and E288_400, experimental pseudo-data replicas, and a PDF-member overlay in the TMD reconstruction. The primary production object is a smooth and audited $b_T$-space TMD ensemble. A regularized finite-$b_T$ Hankel transform gives a companion $k_T$-space representation over $0 \le k_T \le 4$ GeV.

The note also explains what is meant by the accuracy claim. The claim is not that this is a complete global-DY uncertainty budget. Rather, the fixed-target result is accurate within a precisely specified scheme: the N$^3$LL resummed backend algebra closes at numerical precision, the fixed-target data conventions and one known bad row have been audited, experimental pseudo-data generation has been verified through the `target_used` column, the 50-replica ensemble passes fit and random-split stability gates, the PDF uncertainty is propagated as an overlay in the TMD reconstruction, and the $k_T$ representation is stable against several finite-$b_T$ regularization choices. Missing components are stated explicitly: PDF-through-refit effects, scale/profile variations, nuclear-model variations, model-form variations, and accelerator/collider DY data.


## Contents

# 1 Executive summary

The present fixed-target DY result is complete enough to serve as the baseline for the next project stage. The finalized scope is:

- data sets: E288_200, E288_300, E288_400, E605, and E772;
- one explicitly corrected row: E288_300:99;
- explicit 15% normalization priors;
- 5% point-to-point (P2P) uncertainty sensitivity on E772 and E288_400;
- v22 full $N^3$LL-resummed perturbative backend (`resum-order n3llp`), including the NLO OPE/hard upgrade and the small-$b$ profile;
- 50 accepted experimental data replicas for the fixed-target fit;
- PDF-member overlay in the final TMD reconstruction;
- primary production result in $b_T$ space;
- regularized finite-$b_T$ Hankel-transform companion in $k_T$ space over $0 \leq k_T \leq 4$ GeV.

The word "complete" in this note should be read with the above scope. We have not solved the full global-DY problem, and we have not produced a theory-envelope uncertainty. What we have completed is a controlled fixed-target extraction in which the perturbative backend, data conventions, pseudo-data sampling, replica stability, PDF overlay, and $b_T/k_T$ representations have all passed explicit numerical gates. Sections 2 and 6 give the detailed basis for this statement.

The most important final pass/fail summaries are collected in Table 1. The outcome is that the fixed-target DY $b_T$-space ensemble and the regularized $k_T$-space companion are both usable. Accelerator/collider data are not yet part of this fit and require a separate v23b intake, covariance, electroweak-process, and bin-integration effort.

Table 1: Final audited status of the fixed-target DY result.

| Component | Status | Key values |
|---|---|---|
| Central v23a p2p5 refit | PASS | $\chi^2_{\text{tot}} = 0.9583$, max dataset $\chi^2 = 1.4252$, max norm pull = 0.2297 |
| N$^3$LL resummed backend | PASS | `resum-order n3llp`; N$^3$LL logarithmic accuracy in the $W$ term with v22 NLO hard/OPE matching |
| 50-rep fixed-PDF data-replica ensemble | PASS | $\chi^2_{95} = 2.1310$, norm-pull$_{95}$ = 2.7074, random-split width $q_{90} = 0.6439$ |
| $b_T$-space data+PDF overlay TMD | PASS | smooth bands; central PDF0 nearly on ensemble median for standard $d$, $x = 0.1$, $Q = 10$ GeV plot |
| Regularized $k_T$ transform | PASS | expb2/expb/taper p90 difference = 0.0309, max difference = 0.0430 over active $k_T \leq 4$ GeV |
| Remaining large-$k_T$ oscillations | diagnostic | worst median dip about $-1.72\%$ of peak; not clipped |
| Full global/accelerator DY | NOT INCLUDED | requires separate data, covariance, EW, bin-integration, and FO/Y validation |

## 2 Meaning of the accuracy claim

This note uses the word "accuracy" in a deliberately operational sense. The result is accurate within the fixed v23a scheme if all of the following layers are under control:

1. the perturbative/backend calculation is algebraically self-consistent and closes against independent implementations at numerical precision where such checks are available;
2. the data table used by the fit has stable column conventions, units, prefactors, row corrections, and uncertainty prescriptions;
3. the central fit describes the selected fixed-target data without large normalization pulls or pathological residuals;
4. pseudo-data replicas genuinely sample the experimental uncertainties used in the training objective;
5. the replica ensemble is stable under split/random-split tests;
6. the exported TMDs are smooth and finite in $b_T$ space;
7. the companion $k_T$ representation is stable against the finite-$b_T$ regularization choices used to define it.

Each of these statements is tested below. The accuracy claim does *not* include accelerator/collider data, full electroweak $Z/W$ treatment, full covariance matrices for all experiments, PDF-through-refit effects, scale/profile variations, nuclear-model variations, or a model-form envelope. Those are future extensions rather than hidden assumptions in the present result.

### 2.1 The perturbative accuracy label: $\mathrm{N^3LL}$

The perturbative accuracy label for this fixed-target extraction should be written explicitly as a $\mathrm{N^3LL}$-resummed TMD extraction. In the run scripts this is the setting

`--resum-order n3llp`.

The code label `n3llp` is the historical backend label for the $\mathrm{N^3LL}$-level resummed kernel used in the v22/v23a work. In the text of this note we use the physics label $\mathrm{N^3LL}$ because the relevant statement is the logarithmic accuracy of the resummed $W$ term.

The schematic $b_T$-space singular contribution has the form

$$W(b_T, Q, x_1, x_2) = H(Q, \mu_Q)\, e^{-S(b_T,Q)} \left[C \otimes f\right]_1 (x_1, \mu_b) \left[C \otimes f\right]_2 (x_2, \mu_b) F_{\rm NP}(x_1, x_2, b_T; \theta), \quad (1)$$

where the Sudakov/Collins–Soper evolution exponent resums the large logarithms generated by the hierarchy between $Q$ and the transverse scale $1/b_T$. Calling the result $\mathrm{N^3LL}$ means that the logarithmic evolution in this resummed $W$ term is kept through next-to-next-to-next-to-leading-logarithmic accuracy in the backend scheme used here. The hard factor and OPE coefficient insertions used in the final v22 kernel are included at strict NLO, and the profile-dependent general-scale OPE and small-$b$ profile are part of the same audited backend.

It is important to say this accurately. The fixed-target result should not be described merely as a neural-network fit with a generic perturbative kernel. A central achievement of this work is that the nonperturbative DNN is fitted on top of a $\mathrm{N^3LL}$-resummed perturbative TMD kernel, rather than absorbing large perturbative logarithms into the network. This separation is central to the philosophy of the extraction: perturbative QCD controls the short-distance logarithmic evolution and matching, while the DNN is reserved for the genuinely nonperturbative transverse structure.

At the same time, the label should not be overextended. In this note $\mathrm{N^3LL}$ refers to the resummed $W$-term accuracy used for the fixed-target TMD extraction. The full matched observable, especially the finite-$Y$ contribution and high-$q_T$ tail relevant for accelerator/collider data, is not yet advertised as a fully externally validated $\mathrm{N^3LL}$ global-DY prediction. A concise and accurate label is therefore:

$\mathrm{N^3LL}$-*resummed fixed-target DY TMD extraction with NLO hard/OPE matching in the* $W$ *term.*

### 2.2 Validation stack

Table 2 summarizes the validation stack. The important point is that the final TMD result is not supported by a single low $\chi^2$ number. It is supported by a chain of independent checks: backend bridge identities, perturbative finite-difference audits, data-convention audits, central-fit metrics, pseudo-data sampling audits, replica stability tests, PDF-overlay reconstruction checks, and $k_T$ regularization comparisons.

Table 2: Operational validation stack for the fixed-target result. The final claim is limited to this stack.

| Layer | What was tested | Representative outcome |
|---|---|---|
| Born/luminosity bridge | independent luminosity and Bessel identities | relative closures at $10^{-15}$–$10^{-16}$ level in the bridge audits |
| N$^3$LL resummation | resummed $W$-term logarithmic accuracy through `resum-order n3llp` | short-distance evolution kept at N$^3$LL while the DNN only fits $F_{\rm NP}$ |
| NLO OPE/hard backend | strict-linear NLO insertion and hard-factor combination | finite values, positivity after profiling, beyond-NLO diagnostics at the percent level |
| Small-$b$ profile | behavior of OPE logs in the $q$-cap region | raw negative ratios removed; profiled minimum strict ratio 1.076 |
| Data convention | fixed-target prefactors, row keys, units, and uncertainties | E605/E772 consistent with $A$/PreFactor; E288_300:99 corrected |
| Central fit | fixed-target p2p5 central refit | $\chi^2_{\rm tot} = 0.9583$, max dataset $\chi^2 = 1.4252$, max norm pull 0.2297 |
| Pseudo-data sampling | rowwise variation of `target_used` | median target spread $1.41\,\sigma_{\rm used}$; norm-stripped shape spread $1.04\,\sigma_{\rm used}$ |
| Replica convergence | 50-replica fit/norm/split tests | $\chi^2_{95} = 2.1310$, norm-pull$_{95}$ = 2.7074, random-split pass |
| PDF overlay | perturbative/OPE TMD reconstructed with PDF members 1–50 | exp+PDF overlay bands produced for the final $b_T$ grids |
| $k_T$ representation | expb2/expb/taper regularization comparison | p90 regularization difference 0.0309 over active $k_T \leq 4$ GeV |

### 2.3 What is and is not being benchmarked

Several of the audits are numerical closure tests, not phenomenological validation against an independent experiment. For example, the Born-luminosity and OPE bridge checks show that the new kernel is internally consistent with the intended normalization and leg conventions. The MCFM/DYTurbo one-point exercise is an external finite-order normalization diagnostic, but it is not yet a full collider-scale $Y$-term validation. Likewise, the $k_T$ closure integral is band-limited and regularization-dependent; it is used as a diagnostic rather than as a hard normalization theorem for the finite grid.

The phenomenological accuracy statement is therefore narrower and stronger: within the selected fixed-target data, the specified N$^3$LL v22 backend, the p2p5 uncertainty prescription, the stated normalization treatment, and the exp+PDF overlay TMD reconstruction, the result passes all gates we designed for a fixed-target baseline. This is the reason it is suitable as the baseline before beginning the accelerator-data project.

## 3 N$^3$LL perturbative/backend upgrades leading to v22

The fixed-target result uses the v22 full backend, not the older v19/v21 perturbative kernel. The backend work proceeded through several audited steps.

The backend is not a phenomenological black box. It is the N$^3$LL-resummed CSS/TMD perturbative kernel used to compute the singular $W$ term before multiplication by the fitted nonperturbative factor. The trainer precomputes this perturbative kernel on the selected data rows and then learns only the smooth $F_{\text{NP}}$ correction. This is why the N$^3$LL label belongs in the title, abstract, and summary of the result.

### 3.1 Why N$^3$LL matters in this extraction

Fixed-target DY at low transverse momentum contains logarithms of the form $\ln(Q^2 b_T^2)$ that would spoil a fixed-order treatment in the small-$q_T$ region. The N$^3$LL backend resums these logarithms in the Sudakov/Collins–Soper evolution part of $W(b_T, Q)$. The DNN therefore does not have to mimic missing perturbative logarithms. Instead it fits the residual nonperturbative transverse structure after the N$^3$LL perturbative evolution, hard factor, OPE matching, profile choice, and PDF convolution have been applied.

This distinction is central to the accuracy claim. A low $\chi^2$ by itself would not be impressive if the DNN were compensating for a low-order or inconsistent perturbative kernel. The v22/v23a result is stronger: the fitted $F_{\text{NP}}$ sits on a N$^3$LL-resummed kernel whose bridge identities, NLO hard/OPE insertion, general-scale behavior, and small-$b$ profile have been audited separately.

### 3.2 Canonical-window NLO OPE insertion

The first step was to verify the NLO operator-product-expansion (OPE) insertion in a canonical region where the bridge to the legacy Born luminosity and $W$ kernel could be checked directly. The canonical audit found finite values and closed all bridge checks. In the representative fixed-target points,

$$\frac{\delta W_{qq}}{W_{\text{Born}}} \in [-0.0781, -0.0682], \tag{2}$$

$$\frac{\delta W_{qg}}{W_{\text{Born}}} \in [0.0042, 0.0203], \tag{3}$$

$$\frac{\delta W_{\text{OPE}}}{W_{\text{Born}}} \in [-0.0720, -0.0516]. \tag{4}$$

The maximum difference between the naive NLO product and strict linear NLO treatment was about $1.3 \times 10^{-3}$ relative to Born, so the strict-linear implementation was used.

### 3.3 Hard factor plus OPE

The hard correction was then added at strict NLO. The hard factor is positive and partially cancels the negative OPE correction. The representative ranges were

$$\frac{\delta W_{\text{hard}}}{W_{\text{Born}}} \in [0.1201, 0.1424], \tag{5}$$

$$\frac{\delta W_{\text{OPE}}}{W_{\text{Born}}} \in [-0.0720, -0.0516], \tag{6}$$

$$\frac{\delta W_{\text{hard+OPE}}}{W_{\text{Born}}} \in [0.0522, 0.0908]. \tag{7}$$

The maximum beyond-NLO term in the diagnostic was about 0.0081, which was small enough for the strict NLO kernel to pass the audit.

### 3.4 General-scale OPE and profile support

The v22 backend had to support a nontrivial scale profile rather than only the canonical $\mu_b$ choice. A profile-support audit showed that most bare perturbative Bessel weight lives in the $\mu$-floor region for the conservative no-$F_{\text{NP}}$ diagnostic. This made a general-scale OPE implementation operationally necessary. The code path was therefore upgraded with a general-scale CSS2 OPE implementation and corresponding smoke checks.

The important point is conceptual: the profile-support audit used bare perturbative support. In the final fits the nonperturbative factor damps large $b_T$, so the audit was a conservative code-development diagnostic rather than a physics acceptance criterion.

### 3.5 Small-$b$ profile

The raw general-scale OPE logs produced unstable behavior at very small $b$. A smooth small-$b$ profile was introduced for the perturbative coordinate entering the OPE logarithms. The raw profile had negative strict ratios in the $q$-cap region, with very large beyond-NLO diagnostics, but the profiled version restored positivity and kept beyond-NLO terms at the percent level. The final small-$b$ profile audit passed with:

- raw global strict-ratio minimum: $-12.62$;
- profiled global strict-ratio minimum: 1.076;
- raw nonpositive total: 4;
- profiled nonpositive total: 0;
- profiled maximum beyond-NLO diagnostic: about 0.0108.

This profile is a theory/profile choice to be varied later; it was not fitted to data.

### 3.6 Standalone and integrated v22 kernel

The final standalone $W$-kernel audit gave, for four reference rows,

$$\frac{W_{\text{strict}}}{W_{\text{old}}} \in [1.1415, 1.2046], \tag{8}$$

$$|\Delta_{\text{beyond NLO}}|_{\max} \simeq 0.0108, \tag{9}$$

and passed positivity and bridge checks. The full backend integration audit then confirmed that the full v22 backend reproduced the standalone reference and that the scheme-$Y$ flag behaved as expected. The full $W$ ratio relative to the old kernel was roughly 1.14–1.21 on the reference rows.

# 4 External finite-order/Y diagnostics

A singular-subtraction audit initially indicated a scheme-consistency mismatch and requested a $Y$ rebuild. After aligning the singular subtraction, the same audit gave zero current-vs-v22 overlap difference and set `Y_REBUILD_REQUIRED=False`. This means the internal scheme decomposition is algebraically consistent.

A separate MCFM/DYTurbo one-point regression was used as an external normalization diagnostic. The documented primary target was the E288_400:80 onepoint bridgecut case, with a local MCFM integral of 0.00145969 fb. The backend itself is not in fb units, so this target was used to lock a local finite-order conversion through `FO_real_repaired`, not as a direct $W + Y$ pass/fail. The final note should keep this caveat: the NLO finite-tail/Y path is scheme-consistent and useful, but a complete external collider-scale $Y$ validation is still future work.

# 5 Fixed-target data construction and corrections

## 5.1 Initial issue: raw global-DY column convention

The first raw global-DY intake incorrectly treated new raw-file columns too directly. In the new files the columns `A` and `dA` are the tabulated cross section-like entries, while the existing fixed-target data convention uses `CS` and `error`, with `PreFactor` where applicable. Directly using `A` without the proper prefactor would silently change the E288/E605/E772 normalization. This was caught by comparing the v23a candidate files to the current `Data/` files.

The verified v23a fixed-target table was therefore built from the existing `Data/` source files, not the raw direct-A candidate. The source-validation audit confirmed that E605 and E772 are consistent with `A/PreFactor` and `dA/PreFactor` to numerical precision. Some E288 rows had known source inconsistencies in the prefactor columns, but only one matched-row inconsistency was truly suspect at the working tolerance.

## 5.2 Correcting E288_300:99

The one problematic matched row was E288_300:99. It had

$$\texttt{CS} = 0.353005714, \tag{10}$$

$$\texttt{A/PreFactor} = 0.0353005714, \tag{11}$$

so the explicit `CS` value was a factor of ten above the prefactor-consistent value. Neighboring rows in the same $Q$ and $q_T$ sequence were smooth in the prefactor-consistent convention. The row was therefore corrected in the v23a working table, and the correction is tracked as a provenance/data-cleaning decision.

## 5.3 Matched/training rows

The verified source table contains 540 matched rows with $q_T/Q \leq 0.5$ over E288/E605/E772, and 332 TMD-core rows with $q_T/Q \leq 0.2$. The trainer's final selection for the v23a p2p5 fixed-target fit used 418 rows, distributed as shown in Table 4. The difference between the verified matched table and the trainer selection is due to the training-mode cuts and internal selection rules.

### 5.4 P2P sensitivity on E772 and E288_400

The 20-replica v23a pilots without additional P2P uncertainty were stressed by extremely small reported point-to-point errors in E772 and selected E288_400 bins. Several dominant stress rows had relative errors as low as roughly 2–3% in the raw diagonal uncertainty, which was implausibly restrictive for the fixed-target setting once only diagonal information was retained. A controlled sensitivity branch was therefore created with an additional 5% P2P uncertainty on E772 and E288_400, while retaining explicit 15% normalization priors.

The p2p5 central run verified that this additional P2P component actually entered `sigma_used`:

- E772 minimum `sigma_used/CS`: 0.05599;
- E288_400 minimum `sigma_used/CS`: 0.05311;
- E605 minimum `sigma_used/CS`: 0.07251.

This branch is still named as a sensitivity treatment because the 5% P2P component should be replaced by documented covariance/P2P information if available.

## 6 DNN architecture and fitting philosophy

The DNN is deliberately not asked to learn the full cross section. The hard/short-distance part of the calculation is supplied by the N$^3$LL-resummed v22 backend. The neural network only supplies the residual nonperturbative factor $F_{\rm NP}$, subject to positivity, smoothness, monotonicity, and anchoring constraints. This is the defining architecture choice of the analysis: perturbative QCD is kept in the backend, and the DNN is a controlled nonperturbative interpolator.

The neural network in this analysis is not an end-to-end surrogate for the cross section. It learns only the nonperturbative $b_T$-space factor multiplying a fixed perturbative kernel. The perturbative luminosities, coefficient functions, Sudakov evolution, Bessel weights, prefactors, target/isospin treatment, and $Y$ term are supplied by the backend or by precomputed grids. This separation is central to the philosophy of the fit: the DNN should learn the minimal smooth nonperturbative information needed by the data, while perturbative QCD and the collinear PDFs remain explicit and auditable.

### 6.1 Prediction model

For a data row $i$, the backend supplies a precomputed kernel matrix $K_{i\ell}$ on a $b_T$ grid. This kernel already contains the integration weights, the factor $b_T$, the Bessel function $J_0(q_{T,i}b_T)$, and the perturbative $W$ kernel in the chosen cross-section convention. The model prediction before the fitted data-set normalization nuisance is schematically

$$T_i(\theta) = Y_i + \sum_{\ell} K_{i\ell}\, F_{\rm NP}(x_{1i}, b_{T,\ell}; \theta)\, F_{\rm NP}(x_{2i}, b_{T,\ell}; \theta), \tag{12}$$

where the fixed-target production runs use no additional learned Collins–Soper kernel. A fitted data-set normalization nuisance $n_{d(i)}$ then gives

$$\widehat{T}_i(\theta, n) = n_{d(i)} T_i(\theta). \tag{13}$$

The fit minimizes a scaled mean-square residual plus nuisance and regularization terms,

$$\mathcal{L} = \frac{1}{N}\sum_i \left[\frac{\widehat{T}_i - t_i}{\sigma_i}\right]^2 + \lambda_N \frac{1}{N}\sum_d \left(\frac{n_d - 1}{\Delta_d}\right)^2 + \lambda_{\rm anchor} \left\langle (\log F_{\rm NP} - \log F_{\rm NP,ref})^2 \right\rangle_{x,b} + \cdots . \tag{14}$$

Here $t_i$ is the training target. For central fits $t_i$ is the corrected cross section. For replica fits $t_i$ is `target_used`, the explicit pseudo-data target sampled from the experimental errors and the data-set normalization draws. The ellipsis denotes optional smoothness and monotonicity scaffolds; for the final replica protocol the dominant stabilizer is the log-$F_{\rm NP}$ anchor.

### 6.2 The FiLM nonperturbative factor

The fitted object is a shared single-TMD nonperturbative factor $F_{\rm NP}(x, b_T)$. In the direct form it is parameterized as

$$F_{\rm NP}(x, b_T) = \exp[-b_T^2 A_\theta(x, b_T)], \qquad A_\theta \geq 0, \tag{15}$$

which enforces $F_{\rm NP}(x, 0) = 1$ and positive damping. The code also supports a monotone cumulative form,

$$F_{\rm NP}(x, b_T) = \exp\left[-\int_0^{b_T} d\beta\, 2\beta\, A_\theta(x, \beta)\right], \qquad A_\theta \geq 0, \tag{16}$$

which guarantees nonincreasing behavior on the supplied $b_T$ grid. The final exported grids pass the monotonicity and endpoint-tail audits.

The scalar field $A_\theta(x, b_T)$ is represented by a FiLM-modulated residual network. The radial input features are

$$r(b_T) = \left(b_T, b_T^2, \sqrt{b_T + \epsilon}, \log(1 + b_T)\right), \tag{17}$$

while the conditioning features are

$$c(x) = \left(x, \log\frac{x}{1-x}\right). \tag{18}$$

The radial stream is passed through residual FiLM blocks. Each block uses the $x$-conditioner to produce a multiplicative scale and additive shift for the radial features before applying the residual nonlinearity. The production architecture used a compact network, with width 48, conditioning width 32, and three FiLM blocks. The final head is initialized to a Gaussian-like damping baseline so that the central solution begins close to a simple positive $F_{\rm NP}$ rather than an arbitrary function.

### 6.3 Why this architecture was chosen

The fixed-target data constrain a universal smooth damping function, not a completely arbitrary high-dimensional function of flavor, $x$, $Q$, and $b_T$. The architecture therefore deliberately imposes several physics biases:

- $F_{\rm NP}(x, 0) = 1$ so that the small-$b_T$ normalization is controlled by the perturbative OPE and PDFs;
- $F_{\rm NP} > 0$ so that the nonperturbative factor does not introduce artificial sign changes in $b_T$ space;
- smooth radial features rather than a bin-by-bin free function;
- a shared flavor-independent $F_{\rm NP}$, with flavor dependence entering through the collinear PDFs and OPE coefficients;
- optional monotone/cumulative damping and a smooth late-$b_T$ damping floor;
- a function-space log-$F_{\rm NP}$ trust region for pseudo-data replicas.

These biases are not cosmetic. Without them the fixed-target-only problem is underconstrained: large-$b_T$ oscillations, normalization degeneracies, and overfitting of individual bins can produce equally good cross-section fits but unstable TMDs. The DNN is therefore used as a controlled non-parametric interpolator for a physically constrained nonperturbative factor, not as an unconstrained black box.

### 6.4 Central fit versus replica fits

The central fit is allowed to find the best smooth $F_{\rm NP}$ for the corrected fixed-target table. The replica fits are then initialized from the central checkpoint. For the final 50-replica protocol, the replica training is intentionally more conservative: the training scope is restricted mainly to the final nonperturbative head, and a log-$F_{\rm NP}$ anchor with strength $\lambda_{\log F} = 3$ keeps the replica functions in a trust region around the central solution. This is why the final $b_T$-space uncertainty band is smooth and modest even though the pseudo-data cross sections fluctuate at the intended experimental level.

This design choice explains the interpretation of the uncertainty band. The band is a stable fixed-target experimental+PDF overlay band within the specified model class. It is not a model-form uncertainty envelope. Varying the anchor strength, relaxing the training scope, changing the $F_{\rm NP}$ ansatz, or including scale/profile variations would enlarge the uncertainty budget and belongs to a later systematic-study layer.

## 7 Central v23a refit

The central v23a fixed-target refit using the corrected row, 15% normalization priors, and p2p5 E772/E288_400 treatment passed cleanly. The central metrics are in Table 3.

Table 3: Central v23a p2p5 fixed-target refit summary.

| Metric | Value |
|---|---:|
| Number of prediction rows | 418 |
| Total $\chi^2$-like value | 0.9582974964 |
| Maximum dataset $\chi^2$-like value | 1.4252333189 |
| Maximum absolute normalization pull | 0.2296765596 |
| Core prediction columns finite | True |
| Central refit basic pass | True |

Table 4: Per-dataset central refit metrics for the p2p5 branch.

| Dataset | Rows | $\chi^2$-like | median \|pull\| | median pred/data |
|---|---:|---:|---:|---:|
| E288_200 | 64 | 0.618053 | 0.396916 | 0.997194 |
| E288_300 | 80 | 0.708016 | 0.547287 | 0.976842 |
| E288_400 | 113 | 0.816300 | 0.493654 | 1.011486 |
| E605 | 74 | 1.425233 | 0.842772 | 0.917378 |
| E772 | 87 | 1.226006 | 0.696276 | 0.941540 |

The central refit lowered the replica stress without spoiling the fixed-target description. The largest remaining central $\chi^2$-like contribution was E605, but it was still well below the central pass threshold.

# 8 Experimental pseudo-data replicas and normalization treatment

## 8.1 Replica generation

The trainer samples experimental pseudo-data through a target column written as `target_used`. The generator draws one multiplicative normalization shift per dataset and then adds rowwise Gaussian noise using the uncorrelated/P2P training uncertainty. The important audit outcome is that `target_used`, not `target`, is the sampled training target.

The 50-replica target audit found:

- `target_used` rowwise standard deviation is nonzero for all 418 rows;
- median `target_std/sigma_used`: 1.405681;
- median `target_std/CS`: 0.226483;
- the mean sampled target is centered on the data: mean `(target_mean-CS)/sigma_used` $\simeq -0.007$;
- dataset-level median `target/CS` standard deviations are about 15% for all five datasets.

After stripping an estimated dataset-level normalization draw, the rowwise shape fluctuation was also correct:

$$\text{median}\left(\frac{\sigma_{\text{shape}}}{\sigma_{\text{used}}}\right) = 1.037652, \tag{19}$$

$$\text{mean}\left(\frac{\sigma_{\text{shape}}}{\sigma_{\text{used}}}\right) = 1.042339. \tag{20}$$

Thus the experimental pseudo-data sampling is working. The narrowness of the $b_T$-space TMD band is not caused by missing rowwise experimental fluctuations.

## 8.2 Why the TMD band is narrower than observable-space errors

The observable-space prediction spread is roughly 15% in relative halfwidth, consistent with the normalization-draw and experimental-replica protocol. The TMDPDF band is much narrower because it is a universal, profiled-nuisance, smooth, flavor-independent $F_{\text{NP}}$ shape multiplied by a perturbative/PDF factor. Dataset normalization nuisance parameters absorb much of the correlated normalization variation. Many rowwise fluctuations average out in a smooth universal nonperturbative function. The current $\lambda_{\log F} = 3$ anchor further stabilizes the shape.

This interpretation is important for labels. The fixed-PDF data-replica band alone should not be called a full experimental TMD uncertainty. After adding the PDF overlay, the correct label is “experimental+PDF overlay,” with the explicit caveat that the PDF uncertainty has not been propagated through a refit.

## 8.3 Fifty-replica fit and split stability

The 50-replica fixed-PDF data-replica ensemble passed the q95 and random-split gates, as shown in Table 5. This was necessary because the 20-replica pilots had acceptable fit/norm values but unstable width split diagnostics.

Table 5: Final 50-replica data-replica stability summary.

| Metric | Value |
|---|---|
| Number of replicas | 50 |
| $\chi^2$ median | 1.8586031651 |
| $\chi^2_{95}$ | 2.1310145686 |
| $\chi^2_{\max}$ | 2.1866746954 |
| Norm-pull$_{95}$ | 2.7074333429 |
| Norm-pull$_{\max}$ | 3.0173649788 |
| Random-split width $q_{90}$ | 0.6439466887 |
| Random-split center $q_{90}$ | 0.0217491809 |
| Fit pass | True |
| Norm pass | True |
| Band pass | True |
| Random-split pass | True |

## 9 PDF overlay uncertainty

The fixed-target data-replica fits were initially built with the central PDF member, `NNPDF40_nnlo_as_01180` member 0. This means the collinear PDF/OPE/Sudakov factor was deterministic in the original 50-replica $b_T$-space TMD band. To include a first realistic PDF uncertainty without spending multiple days rebuilding fixed-target cross-section caches and retraining for every PDF member, a PDF overlay was constructed.

For replica $r$ with fitted nonperturbative function $F_{\rm NP}^{(r)}$ and assigned PDF member $m_r$, the overlay TMD is

$$\widetilde{f}_q^{(r,m_r)}(x, b_T; Q) = \left[C_{q\leftarrow j} \otimes f_j^{(m_r)}\right](x; \mu_b) \exp\left[-\frac{1}{2}S^{(m_r)}(b_T, Q)\right] F_{\rm NP}^{(r)}(x, b_T). \tag{21}$$

The final overlay used 50 experimental data-replica $F_{\rm NP}$ fits paired with PDF members 1–50. This is not the same as PDF-through-refit, because $F_{\rm NP}^{(r)}$ was not refit with the PDF member $m_r$. It is nevertheless the correct next level of TMD uncertainty for a note: the TMD reconstruction itself now varies over both experimental data replicas and PDF members.

## 10 $b_T$-space TMD result

The $b_T$-space TMD is the primary production object. In the final grid the hard factor is excluded from the single-TMD object, and the definition is

$$\widetilde{f}_q(x, b_T; Q) = \left[C_{q\leftarrow j} \otimes f_j\right](x; \mu_b, \zeta_b = \mu_b^2) \exp\left[-\frac{1}{2}S(b_T, Q)\right] F_{\rm NP}(x, b_T). \tag{22}$$

The fitted $F_{\rm NP}$ is flavor independent in this model. The flavor dependence of the TMDPDF enters through the collinear PDFs and the OPE coefficient functions.

The standard $b_T$-space sanity plot is the $d$-quark, $x = 0.1$, $Q = 10$ GeV curve. It is shown in Fig. 1. The plot shows a smooth $b_T$ dependence, a visible but modest 68% experimental+PDF overlay band, selected thin replicas clustered around the median, and a central PDF0 curve nearly on top of the ensemble median.

Missing figure file:

`bspace-result.png`

Standard $b_T$-space figure: copy or generate `bspace-result.png` in the note directory.

Figure 1: Standard $b_T$-space TMDPDF example: $d$ quark, $x = 0.1$, $Q = 10$ GeV. The shaded band is the 68% experimental+PDF overlay interval. The dashed black curve is the central fit with PDF member 0. The thin gray curves are selected replica curves for visual diagnostics.

The caption used for the note/paper should emphasize that this is an overlay uncertainty, not a PDF-through-refit uncertainty:

> The shaded band is the 68% interval over the fixed-target DY data-replica ensemble with a PDF-member overlay in the perturbative/OPE TMD factor. The dashed curve is the central fit with PDF member 0. This band includes experimental pseudo-data variation and PDF-member variation in the TMD reconstruction, but not PDF-through-refit, profile/scale, nuclear-model, or functional-form variations.

# 11 Regularized $k_T$-space companion

## 11.1 Transform convention

The $k_T$-space companion is obtained by a finite-$b_T$ regularized Hankel transform,

$$f_q(x, k_T; Q) = \frac{1}{2\pi} \int_0^\infty db_T\, b_T\, J_0(k_T b_T)\, \widetilde{f}_q(x, b_T; Q). \tag{23}$$

Formally,

$$\widetilde{f}_q(x, b_T = 0; Q) = 2\pi \int_0^\infty dk_T\, k_T\, f_q(x, k_T; Q). \tag{24}$$

Because the numerical $b_T$ grid is finite and the high-$k_T$ tail is not a perturbative fixed-order calculation, Eq. (23) is implemented as a regularized finite-$b_T$ representation. The default prescription is:

- large-$b_T$ continuation: $\exp(-a b_T^2)$ fitted from the tail (`expb2`);
- transform range: $b_T \leq 24$ GeV$^{-1}$;
- endpoint taper starts at $0.92\, b_T^{\max}$;
- final quoted range: $0 \leq k_T \leq 4$ GeV;
- negative lobes are retained and audited, not clipped.

### 11.2 Regularization stability

Three tail/regularization modes were compared: `expb2`, `expb`, and direct `taper`. The median curves are stable over the active $k_T \leq 4$ GeV region:

$$\max\left[\Delta_{90}^{\text{rel}}(\text{tail modes})\right] = 0.03091254598, \tag{25}$$

$$\max\left[\Delta_{\max}^{\text{rel}}(\text{tail modes})\right] = 0.04304417062. \tag{26}$$

The stability comparison passed with `KT_REGULARIZATION_STABILITY_PASS=True`.

The default `expb2` curve audit found:

- all transformed values finite;
- worst median negative dip relative to peak: $-0.0171855$;
- worst negative area fraction: 0.134224;
- maximum active-region 68% relative halfwidth $q_{90}$: 0.759784.

The largest relative bands occur mainly for high-$x$ sea-quark curves where the absolute TMD is tiny. For example, $\bar{d}$, $x = 0.5$, $Q = 5$ GeV has a very small peak and therefore a large relative band. Main valence/moderate-$x$ curves have much smaller relative bands.

### 11.3 Status of $k_T$ representation

The regularized $k_T$ companion is now finalized with the following status:

> PASS as a regularized finite-$b_T$ Hankel-transform companion over $0 \leq k_T \leq 4$ GeV.

This should not be described as a standalone high-$k_T$ perturbative-tail prediction. It is the $k_T$ representation of the fitted $b_T$ ensemble under a stated regularization prescription.

### 11.4 Standard $k_T$ figure

The standard $k_T$ figure should match the $b_T$ figure: $d$ quark, $x = 0.1$, $Q = 10$ GeV, using `ftilde`. The recommended caption is:

> Regularized $k_T$-space companion for the fixed-target DY extraction. The curves are obtained from the fitted $b_T$-space ensemble by a finite-$b_T$ Hankel transform with $\exp(-ab_T^2)$ large-$b_T$ continuation and smooth endpoint taper. The shaded band is the 68% experimental+PDF overlay interval; the dashed curve is the central PDF0 result. The result is shown over the $k_T$ range stable under `expb2/expb/taper` regularization comparisons.

A placeholder for the final $k_T$ plot is shown in Fig. 2. Replace the placeholder with the output of `plot_v23a_traditional_kspace_tmd.py`.

Missing figure file:

`kspace-result.png`

Standard $k_T$-space figure: copy or generate `kspace-result.png` in the note directory.

Figure 2: Standard $k_T$-space TMDPDF companion for the fixed-target Drell–Yan extraction. Shown is the $d$-quark TMDPDF at $x = 0.1$ and $Q = 10$ GeV. The solid curve is the median of the 50-replica experimental+PDF ensemble, the shaded band denotes the 68% uncertainty interval, and the dashed curve is the central (PDF0) fit. The $k_T$-space distribution is obtained from the fitted $b_T$-space ensemble using the regularized finite-$b_T$ Hankel transform with the default `expb2` prescription.

# 12 Reproducibility commands and artifact paths

## 12.1 Primary fixed-target artifact names

The final fixed-target result should be frozen with explicit names:

```
v23a_lambda3_normpriors15_p2p5_E772_E288400_50rep_DYonly_bspace_sensitivity
v23a_lambda3_normpriors15_p2p5_E772_E288400_50rep_DYonly_kspace_regularized_expPDF_overlay
```

The first is the primary $b_T$-space extraction. The second is the regularized $k_T$ companion.

## 12.2 Key paths

Table 6: Main repository paths used by the final fixed-target result.

| Object | Path |
|---|---|
| Central p2p5 run | `outputs/v23a_fixed_target_lowQ_corrected_central_refit_normpriors15_p2p5_E772_E288400_s303` |
| 50 data-replica root | `replica_pilot_v23a_lambda3_normpriors15_p2p5_E772_E288400_cached_cuda` |
| PDF-overlay $b_T$ bands | `replica_v23a_expPDF_overlay_lambda3_normpriors15_p2p5_50rep/tmd_bspace_bands_expPDF_overlay` |
| Regularized $k_T$ default | `replica_v23a_expPDF_overlay_lambda3_normpriors15_p2p5_50rep/kspace_regularized_expPDF_overlay_expb2` |

| Object | Path |
|---|---|
| Regularization comparison | `replica_v23a_expPDF_overlay_lambda3_normpriors15_p2p5_50rep/kspace_regularized_comparison` |
| Traditional $b_T$ example | `plots/v23a_traditional_expPDF_overlay/ftilde_dataPDF_bands_improved.pdf` |
| Traditional $k_T$ example | `plots/v23a_traditional_kspace_TMDPDF_d_x0p10_Q10.pdf` |

### 12.3 Minimal command skeleton

The exact long paths are listed in Table 6. The reproducibility sequence is:

1. Create the PDF-overlay plan from the existing fixed-PDF 50-replica run directories with `make_v23a_pdf_overlay_plan_from_runs.py`.
2. Construct the PDF-overlay $b_T$-space TMD bands with `construct_v23a_data_pdf_bspace_tmd_bands_v2.py`, using `--pdf-member-source plan`.
3. Make the traditional $b_T$ figure with `plot_v23a_data_pdf_bspace_bands_improved.py`.
4. Build three regularized $k_T$ transforms with `construct_v23a_regularized_kspace_tmd.py`, using `expb2`, `expb`, and `taper` tail modes.
5. Compare the three $k_T$ regularization modes with `compare_v23a_regularized_kspace_modes.py`.
6. Make the traditional $k_T$ figure with `plot_v23a_traditional_kspace_tmd.py`.

The important short options for the production commands are:

| Task | Essential options |
|---|---|
| PDF-overlay plan | `--pdf-members 1-50`, `--member-strategy cycle` |
| $b_T$ grid | `--x-values 0.10 0.20 0.30 0.50`, `--Q-values 5 10` |
| Traditional $b_T$ plot | `--quantity ftilde`, `--flavor d`, `--x 0.10`, `--Q 10` |
| Regularized $k_T$ grids | `--b-transform-max 24`, `--n-b-transform 6001`, `--k-max 4` |
| Regularization comparison | reference mode `expb2`; alternatives `expb`, `taper` |
| Traditional $k_T$ plot | `--quantity ftilde`, `--flavor d`, `--x 0.10`, `--Q 10` |

## 13 Limitations and next steps

The N$^3$LL label is a key strength of the result, but it also has a precise scope. It applies to the resummed $W$ term used for the fixed-target TMD extraction. It should not be read as a claim that every component needed for a full global accelerator-DY matched prediction has already been externally validated at the same level. In particular, the high-$q_T$ finite-$Y$ tail, collider electroweak/fiducial effects, and future scale/profile envelopes remain separate work items.

The current result is a strong fixed-target DY result, but it is not yet a full global DY extraction and not yet a complete theory/model-form uncertainty budget. The remaining limitations are:

- PDF uncertainty is included as an overlay in TMD reconstruction, not through refitting $F_{\mathrm{NP}}$ for each PDF member;
- scale/profile variations are not included in the uncertainty band;
- nuclear/isospin model variations are not included;
- the DNN architecture is intentionally constrained: $F_{\mathrm{NP}}$ is flavor independent, smooth, positive, and anchored in replica fits;

- the 5% P2P treatment for E772 and E288_400 is a sensitivity prescription, not a documented full covariance;
- the $k_T$ result is a regularized finite-$b_T$ representation, not a high-$k_T$ fixed-order tail;
- accelerator/collider data are not included.

The next project stage is v23b accelerator data incorporation. That will require dataset-by-dataset unit and covariance locking, electroweak/Z/W process support, bin integration over $q_T$, rapidity, and mass windows, and external fixed-order/Y validation at collider kinematics.

# A Recommended GitHub README bullets

A concise GitHub summary could read:

- Completed v23a fixed-target DY TMD extraction with E288, E605, and E772.
- Accuracy claim is operational: backend, data convention, pseudo-data sampling, replica convergence, PDF overlay, and $k_T$ regularization all pass explicit gates.
- Uses corrected E288_300:99, 15% normalization priors, and a 5% P2P sensitivity on E772/E288_400.
- v22 full backend includes NLO OPE/hard corrections and a small-$b$ OPE profile.
- Central p2p5 refit passes with $\chi^2_{\text{tot}} = 0.9583$ and max norm pull 0.2297.
- 50-replica fixed-target data-replica ensemble passes q95 and random-split gates.
- Experimental pseudo-data sampling verified through `target_used`; norm-stripped shape fluctuations have median $\sigma_{\text{shape}}/\sigma_{\text{used}} = 1.04$.
- TMD bands include a PDF-member overlay in the perturbative/OPE TMD reconstruction.
- Primary result is $b_T$ space; $k_T$ companion is a regularized finite-$b_T$ Hankel transform stable over $0 \leq k_T \leq 4$ GeV.
- Accelerator/collider data are the next project and are not part of this fixed-target result.

# B Short note-to-self on terminology

Use "experimental+PDF overlay" for the final plotted band. Do not use "full experimental+PDF uncertainty" without qualification, because PDF-through-refit and profile/scale/model variations are not included. Use "regularized $k_T$ companion" rather than "production high-$k_T$ TMD tail."

### Terminology update: always state the perturbative order

In talks, notes, and GitHub summaries, the preferred short label is:

N$^3$LL **fixed-target DY TMD extraction with experimental+PDF overlay uncertainty.**

If more detail is needed, use:

N$^3$LL-resummed $W$ term, NLO hard/OPE matching, fixed v22 profile, 50 experimental replicas, and PDF-member overlay in the TMD reconstruction.

This avoids the misleading impression that the main technical advance was only the DNN. The main advance is the combination of a N$^3$LL perturbative TMD kernel with a constrained neural nonperturbative factor and a reproducible fixed-target uncertainty workflow.